Harvard relied on endowment earnings to meet a large share of university expenses. In its fiscal year ending June 30, 2008, more than one-third of operating revenue came from endowment income. For some of the university's individual departments, the proportion was even higher: the Radcliffe Institute for Advanced Study derived 83% of its revenue from the endowment, the Divinity School 71%, and the Faculty of Arts and Sciences 52%.

Harvard Management Company (HMC), the funds manager of Harvard's endowment, was one of the early adopters of the *endowment model*, which recommends that long-term investors hold lots of illiquid, alternative assets, especially private equity and hedge funds. Advocated by David Swensen in his influential book, *Pioneering Portfolio Management*, the endowment model was based on the economic concept of diversification originally attributable to Harry Markowitz (1952; see chapter 3). Through diversification, a portfolio of many low-correlated assets has a risk-return trade-off superior to that of conventional portfolios consisting of only stocks and bonds. Swensen went further and advocated holding large proportions of illiquid private equity and hedge funds. Not only were these assets supposed to have low correlations to stocks and bonds, but they potentially carried an illiquidity risk premium.

Swensen argued that in liquid markets, the potential for making excess returns (or "alpha"; see chapter 10) was limited. In these markets, crowded with thousands of active managers vying for an edge, information is freely available and almost everyone has access to it. Illiquid asset markets, like venture capital and private equity, had large potential payoffs for investors who had superior research and management skills. Swensen argued that alpha was not competed away in illiquid assets because most managers have short horizons. University endowments, with their longer horizons, would seem to have an advantage in illiquid assets. Swensen recommended that long-term institutions with sufficient resources who can carefully select expert managers in alternative, illiquid assets could achieve superior risk-adjusted returns.

Dutifully following Swensen's advice, many endowments, including Harvard, loaded up with illiquid assets during the 1990s. In 2008, HMC held 55% of its portfolio in hedge funds, private equity, and real assets. Only 30% was in developed-world equities and fixed income, with the remainder of its portfolio in emerging-market equities and high-yield bonds.

In its desperate need for cash, HMC tried to sell some of its $1.5 billion private equity portfolio, which included marquee names such as Apollo Investment and Bain Capital. But buyers in secondary markets demanded huge discounts. Nina Munk, a journalist writing in *Vanity Fair*, recounts a surreal conversation between the CIO of HMC, Jane Mendillo, and a money manager specializing in alternative investments:[3]

[3] Nina Munk, "Rich Harvard, Poor Harvard," *Vanity Fair*, August 2009.